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Longitudinal associations of cardiovascular-kidney-metabolic syndrome with midlife or late-life mental disorders and dementia, and the mediating role of metabolomic signature

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Abstract

Background: Cardiovascular-Kidney-Metabolic (CKM) syndrome assesses the interconnections among metabolic, kidney, and cardiovascular diseases, rendering significant prognostic value for age-related chronic diseases and mortality. We aimed to investigate the effects of CKM syndrome on transitions between healthy status, mental disorders, and dementia and evaluate the potential mediating role of a CKM-related metabolomic signature in these associations.

Methods: This prospective longitudinal study used UK Biobank data from 375,203 midlife and older adults at baseline and 188,018 with metabolomic information. CKM was staged from 0 to 4. Mental disorders and dementia were identified via ICD-10. Multi-state models analyzed the impact of CKM on transitions from healthy status to mental disorders and dementia. Competing risk (death) models assessed the associations of CKM with specific mental disorders and dementia. Mediation role of CKM-related metabolomic signature was evaluated.

Results: We show that per-stage CKM increase elevates hazards of transitioning from healthy to mental disorders (HR=1.24[1.22-1.26]) and subsequently to dementia (HR=1.38[1.21-1.58]), or directly to dementia (HR=1.27[1.21-1.33]). Worsening CKM stages are associated with bipolar, depressive, and anxiety disorders; whilst only advanced stages (3/4) associated with all dementia types. The CKM metabolomic signature mediates 34.9% and 8.1% of associations of CKM with pre-dementia mental disorders and dementia, respectively.

Conclusions: CKM syndrome is associated with pre-dementia mental disorders and dementia, emphasizing the need for regular monitoring and early intervention to manage CKM progression and reduce geriatric neuropsychiatric disturbances.

Plain language summary

Cardiovascular-kidney-metabolic (CKM) syndrome is a 5-stage health condition that includes heart disease, kidney disease, and diabetes. Such conditions are associated with brain disorders. We examined the relationship between CKM stages with mental disorders and dementia, and further examined the impact of blood chemicals on these

relationships. We found that higher CKM stages are linked to mental disorders, while only advanced stages are linked to dementia. Changes in blood chemicals may partly explain these links. These findings highlight the need for monitoring and targeted interventions to address mood and cognitive impairments among people with CKM syndrome.

Editorial Summary

Zhao et al. assess links between cardiovascular–kidney–metabolic (CKM) syndrome, mental disorders, and dementia, and the mediating role of a CKM-related metabolomic signature. Higher CKM stages are linked to mental disorders, while only advanced stages are linked to dementia. The metabolomic signature partly mediates these associations.

1. Introduction

Dementia patients often exhibit neuropsychiatric symptoms, such as affective or psychosis syndromes, particularly in advanced disease stages¹. Mental disorders, including psychotic disorders, bipolar disorders, depressive disorders, and anxiety disorders, are increasingly prevalent in aging populations and have been associated with a substantially elevated risk of dementia in several recent large population-based studies²⁻⁸. For example, depression has been linked to more than a 50% increased risk of dementia, while psychotic disorders are associated with an even greater risk of subsequent dementia. In addition, accumulating evidence indicates that cognitive deficits in individuals with mental disorders not only contribute to greater symptom severity but may also persist after clinical recovery⁹. However, the directionality of the association between mental disorders and dementia remains unclear. Dementia-related pathology may increase the risk of mental disorders either before dementia onset, manifesting as prodromal neuropsychiatric symptoms, or after a dementia diagnosis^{2,10}. Given the close and dynamic interplay between mental disorders and dementia, examining transitions among healthy status, mental disorders, and dementia may provide a more comprehensive understanding of disease development. Identifying risk factors associated with these transitions could inform strategies to reduce the neuropsychiatric burden in aging societies¹¹.

Cardiovascular, kidney, and metabolic diseases are shared independent risk factors for both mental disorders and dementia¹²⁻¹⁷. Kidney and metabolic comorbidities can induce inflammation, oxidative stress, and alterations in peripheral and brain lipid metabolism, whilst vascular diseases may lead to cardiovascular damage, such as white matter dysfunction^{13,18}. Furthermore, cardiovascular, kidney, and metabolic diseases are interrelated at the pathophysiological level and often co-occur^{19,20}. In 2023, the American Heart Association (AHA) introduced the term Cardiovascular-Kidney-Metabolic (CKM) syndrome as characterized by pathological interactions among metabolic risk factors, chronic kidney disease (CKD), and cardiovascular dysfunction²¹. CKM syndrome contributes to multi-organ dysfunction²² and is associated with high premature morbidity and mortality²³⁻²⁵. Based on risk factors and established disease,

the AHA delineates 5 CKM stages, ranging from 0 (no risk factors) to 4 (clinical/established cardiovascular disease, (CVD)), each with corresponding preventive and therapeutic recommendations aimed at improving CKM-related health outcomes. Hence, this conceptual framework of CKM syndrome not only offers a novel perspective for exploring a comprehensive understanding of the connections among CKM, cognitive impairment and neuropsychiatric disorders, which could not only facilitate the elucidation of pathophysiological mechanisms underpinning age-related neuropsychiatric and cognitive impairments, but also pave the way towards developing effective prevention and management strategies for dementia.

CKM syndrome encompasses a broad spectrum of metabolic dysfunctions and may therefore be associated with distinct circulating metabolomic profiles. Previous studies have shown that circulating metabolites are closely linked to nervous system function and the risk of neuropsychiatric disorders^{26,27}. Several serum metabolite biomarkers, including eicosanoids, lipidomic profiles, and energy-related metabolites, have been identified as predictors of mental disorders such as major depressive disorder, bipolar disorder, and schizophrenia^{28,29}. In addition, disrupted fatty acid metabolism has been implicated in the pathogenesis of dementia³⁰. Together, this evidence highlights the importance of investigating whether CKM-related circulating metabolomic profiles modulate the associations between CKM syndrome and psychiatric disorders and dementia, which may provide insight into the underlying pathological mechanisms.

Using data from the UK Biobank prospective cohort, this study examines the associations between CKM syndrome, mental disorders, and dementia in midlife and older adults. We investigated the differential effects of CKM stages on transitions between healthy status, mental disorders, and dementia, and assessed the associations of CKM stages with specific mental disorders and dementia while accounting for the competing risk of death. We also evaluated the potential mediating role of a CKM-related metabolomic signature in these associations. We find that increasing CKM stages are associated with a higher risk of pre-dementia mental disorders, whereas only advanced CKM stages (stages 3/4) are associated with dementia. The CKM-related metabolomic signature partially mediates the associations between CKM syndrome and

both mental disorders and dementia. These findings inform stage-specific monitoring and targeted interventions to address mood and cognitive impairments among individuals with CKM syndrome.

2. Methods

2.1. Data source and study population

The study population was derived from the UKB, a large prospective cohort study comprising over 502,412 participants aged 37–73 years at baseline (2006–2010). Participants were recruited from 22 assessment centers across the United Kingdom. Data collection included touch-screen questionnaires, biological samples, physical measurements, and linked medical or death registry records. The detailed study design and methodology have been described elsewhere³¹. The UKB is approved as a Research Tissue Bank by the North West Multicentre Research Ethics Committee, eliminating the need for separate ethical approvals for individual projects. All participants provided signed informed consent at the baseline assessment. For this study, participants with complete data for defining their baseline CKM stage were included. Individuals with baseline diagnoses of mental disorders, dementia, or other chronic brain disorders (e.g., Parkinson's disease) were excluded (see Supplementary Table 1 for more details). For metabolomics analyses, we included a subgroup of participants with available plasma metabolomics measurements. The study is reported as per the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guideline (see STROBE statement).

This study was conducted in accordance with the Declaration of Helsinki and was approved by the Ethics Committee of the School of Public Health, Zhejiang University.

2.2. Definition of CKM stage

We defined CKM stages in accordance with the AHA Presidential Advisory Statement on CKM Syndrome, adapted to the data available in the UKB²². Relevant components of CKM stage were assessed using body measurements, biochemical testing of blood samples, self-reported medical history, and medical records (Supplementary Data 1).

Specifically, body mass index (BMI) was calculated from height (m) and weight

(kg). Prediabetes was defined by glycated hemoglobin (HbA1c) levels between 5.7% and 6.4%. Hypertriglyceridemia was defined by serum triglycerides (≥ 135 mg/dL). Hypertension was defined by blood pressure measurements (systolic blood pressure [SBP] ≥ 140 mm Hg and/or diastolic blood pressure [DBP] ≥ 90 mm Hg), self-reported hypertension, medical history, and/or the use of antihypertensive medications. Diabetes was defined by HbA1c ($\geq 6.5\%$), self-reported diabetes, medical history, and/or the use of glucose-lowering medications. Metabolic syndrome was defined by the presence of 3 or more of the following: (1) waist circumference ≥ 88 cm for women and ≥ 102 cm for men (≥ 80 cm for women and ≥ 90 cm for men of Asian ancestry); (2) high-density lipoprotein cholesterol (HDL-C) < 40 mg/dL for men and < 50 mg/dL for women; (3) triglycerides ≥ 150 mg/dL; (4) elevated blood pressure (SBP ≥ 130 mm Hg or DBP ≥ 80 mm Hg and/or use of antihypertensive medications); and (5) HbA1c $\geq 5.7\%$. To define CKD, we assessed the estimated glomerular filtration rate (eGFR) based on serum creatinine and Cystatin C using the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) 2012 formula^{32,33}. This formula accounts for age, ethnicity, and sex. CKD status was categorized as: normal (≥ 60 mL/min/1.73 m²), mild to moderately impaired (45–59 mL/min/1.73 m²), moderately impaired (30–44 mL/min/1.73 m²), severe loss of kidney function (15–29 mL/min/1.73 m²), or kidney failure (≤ 15 mL/min/1.73 m²). Since cardiac biomarkers and cardiovascular imaging data were unavailable to identify subclinical CVD, we calculated the Framingham Risk Score (FRS) based on the following variables: age, sex, SBP, HDL-C, total cholesterol, antihypertensive medication use, smoking status (non-smoker vs. current smoker), and diabetes status³⁴. A high predicted 10-year CVD risk ($\geq 20\%$) was used as a proxy for subclinical CVD based on established thresholds. Clinical CVD was defined as a self-reported medical history or medical records indicating any of the following conditions: coronary heart disease, heart failure, stroke, peripheral artery disease, or atrial fibrillation.

Subsequently, CKM stage was defined as follows:

(1) CKM stage 0 (No CKM risk factors): Individuals with normal BMI (< 23 kg/m² for individuals with Asian ethnicity and < 25 kg/m² for other racial and ethnic groups)

and waist circumference (<80 and <90 cm for women and men with Asian race, respectively, and <88 and <102 cm for women and men in all other race and ethnicity categories, respectively), who did not meet criteria for the other stages.

(2) CKM stage 1 (Excess or dysfunctional adiposity): Individuals with an elevated BMI (≥ 23 kg/m² for individuals with Asian race and > 25 kg/m² for all other race and ethnic groups), an elevated waist circumference (≥ 80 and ≥ 90 cm for women and men with Asian race, respectively, and ≥ 88 and ≥ 102 cm for women and men in other race and ethnicity categories, respectively), or prediabetes, without the presence of other metabolic risk factors or CKD.

(3) CKM stage 2 (Metabolic risk factors and CKD): Individuals with hypertriglyceridemia, hypertension, diabetes, CKD or metabolic syndrome.

(4) CKM stage 3 (Subclinical CVD in CKM): Individuals with high predicted 10-year CVD risk (FRS $\geq 20\%$, based on recommended thresholds) or very high-risk CKD (severe loss of kidney function or kidney failure) who also met criteria for stage 1 or stage 2, or CKD.

(5) CKM stage 4 (Clinical CVD in CKM): Individuals with clinical CVD or kidney failure who also met criteria for stage 1, stage 2, or CKD.

2.3. Diagnoses of mental disorders and dementia

Mental disorders were defined as the presence of any diagnosis within four categories: psychotic disorders, bipolar disorders, depressive disorders, and anxiety disorders. These categories were selected for two primary reasons: (1) each has been associated with an increased risk of dementia⁴⁻⁷, and (2) substantial genetic correlations among these phenotypically distinct mental disorders suggest shared underlying genetic architectures^{35,36}. Dementia was defined as all-cause dementia, including Alzheimer's disease and vascular dementia. Both mental disorders and dementia were identified using hospital inpatient records or death register linkage data, where they were recorded as either an underlying or contributory cause. Diagnoses were documented based on the International Classification of Diseases, 10th Revision (ICD-10). The date of diagnosis was determined as the earliest recorded date across all available data sources. Detailed information on the ICD-10 codes used to identify mental disorders and dementia is

provided in Supplementary Data 2 and Supplementary Table 2. In this study, we focused on two paths: Path A represented transition from healthy status to mental disorders and then to dementia. Path B represented transition from healthy status to dementia and then to mental disorders.

2.4. Covariates

Covariates included age (continuous), sex (female or male), education (high or low), socioeconomic status (assessed using the Townsend Deprivation Index³⁷), race (categorized as Asian, White, or Black), alcohol consumption (current, former, or non-drinker), physical activity level (high or low), social connection (high or low), and sleep quality (good or poor). Detailed definitions of all covariates are provided in Supplementary Data 3.

2.5. Plasma metabolomics

Metabolite quantification was performed using EDTA plasma samples from approximately 280,000 UK Biobank participants, analyzed via a high-throughput targeted nuclear magnetic resonance (NMR) metabolomics platform (Nightingale Health Ltd., Finland). This approach quantified 251 predefined metabolites (Supplementary Data 4), ensuring consistent detection across all samples through standardized protocols, including calibration with internal and external quality controls. The NMR spectra were acquired using a Bruker IVDr platform (600 MHz). These metabolites encompass a broad range of metabolic pathways, including fatty acids, fatty acid compositions, and lipoprotein lipids across 14 subclasses, as well as various low-molecular-weight metabolites such as ketone bodies, amino acids, and glycolysis-related metabolites. Detailed protocols for sample collection and the Nightingale NMR methodology have been described previously³⁸. All metabolite values underwent rank-based inverse normal transformation, a two-step process. First, observations were transformed to the probability scale using the empirical cumulative distribution function. Next, the transformed values were converted to Z-scores using the probit function. Our metabolomics analysis included participants with complete data on each metabolite.

2.6. Statistics and reproducibility

Demographic characteristics are presented as mean $\pm$ SD or number (%). Baseline characteristics of participants across different CKM syndrome stages were compared using χ^2 tests or ANOVA, as appropriate. Participants were considered at risk from enrollment until death, loss to follow-up, or 16 December 2022, whichever occurred first.

We initially employed a multi-state model based on Cox proportional hazards regression to investigate the association between CKM stage and the temporal progression from a healthy state to the onset of mental disorders or dementia, via Path A or Path B. This multi-state model, which represents multiple survival curves, is an extension of the competing risks model, allowing exploration of how specific factors influence different phases of a process^{39–41}. Hazard ratios (HRs) for CKM stage were computed for each transition. Models were adjusted for age, sex, race, and education, alcohol consumption, SES, physical activity, social connections, and sleep quality. In the multi-state model, participants who progressed to mental disorders or dementia on the same date were excluded. CKM stage was treated as both a categorical variable (reference = CKM stage 0) and a continuous variable (per 1-unit increase in CKM stage). To assess the robustness of the multi-state model, we additionally fitted a continuous-time multistate Markov model using the R package ‘msm’⁴². In this framework, transitions are characterized by transition intensity functions, which represent the instantaneous rate of moving from one state to another at a given time, conditional on being at risk for that transition^{43,44}. We specified two transient states (mental disorders and dementia) and one absorbing state (death), yielding transition structures consistent with those of the primary multi-state model. CKM syndrome, age, sex, race, education, alcohol consumption, socioeconomic status, physical activity, social connections, and sleep quality were included as covariates, influencing transition intensities under a proportional intensity model. Hazard ratios for CKM syndrome were estimated for all possible transitions.

Next, we conducted Fine-Gray subdistribution hazard models to assess the associations between CKM stage and the incidence of any and different types of mental disorders and dementia, accounting for the competing risks of death⁴⁵. The types of

mental disorders included psychotic disorders, bipolar disorders, depressive disorders, and anxiety disorders. The types of dementia included Alzheimer's disease and vascular dementia. CKM stage was treated as either a categorical variable (with CKM Stage 0-3 as references) or a continuous variable (per 1-unit increase in CKM stage). Two models were conducted. Model 1 was adjusted for age, sex, race, and education, whereas Model 2 additionally included adjustments for alcohol consumption, SES, physical activity, social connections, and sleep quality. Subdistribution hazard ratios (sHRs) were estimated.

Multivariable linear regression models were used to estimate the associations between each metabolite and CKM stages, adjusting for all covariates described above. Bonferroni correction was applied to account for multiple testing and to identify statistically significant associations. Participants were then randomly assigned to training and testing sets in an 8:2 ratio for construction of the CKM metabolomic signature. In the training set, an elastic net regression model with tenfold cross-validation was used to regress CKM stages on the metabolites identified as significant in the multivariable analyses. The resulting model was subsequently applied to the testing set to calculate the metabolomic signature. The signature score was computed as a weighted sum of the selected metabolites, with weights corresponding to the regression coefficients from the elastic net model. Within the training set, signature scores were derived using a leave-one-out cross-validation approach to further reduce the risk of overfitting. To assess the robustness of the metabolomic signature, we calculated correlation coefficients between the signature score and CKM stages in the training set, testing set, and the combined dataset. Finally, to evaluate the potential mediating role of the CKM metabolomic signature in the associations between CKM stages and neuropsychiatric outcomes, we conducted two separate mediation analyses, treating mental disorders and dementia as distinct outcomes and adjusting for the same covariates included in Model 2.

Sensitivity analyses were conducted using multiple imputation with the Markov Chain Monte Carlo method to impute missing data for baseline demographics and covariates, incorporating information from auxiliary variables and generating 5

imputed datasets within the pooled dataset⁴⁶. Estimates derived from each imputed dataset were subsequently combined using Rubin's rules. Additional sensitivity analyses were performed to compare the major findings when only complete cases (i.e., 0% missing baseline data) were included. We also conducted sensitivity analyses further adjusting for self-reported use of cholesterol-lowering medications, antihypertensive medications, and insulin. To address potential reverse causation, main analyses were repeated after excluding participants within 5 and 10 years of follow-up, respectively⁴⁷. R Studio software version 4.1.2 was used, with a statistical significance level set at < 0.05 .

3. Results

3.1. Descriptive analyses

The study flowchart is presented in Figure 1, and the sample characteristics are summarized in Table 1. A total of 113,148 participants were excluded due to a prior diagnosis of dementia, mental disorders, other chronic brain disorders, or incomplete CKM data at baseline. Among the 188,018 participants included at baseline, 9.7% were classified as CKM stage 0 ($N = 18,187$), 7.6% as CKM stage 1 ($N = 14,332$), 57.6% as CKM stage 2 ($N = 108,260$), 18.5% as CKM stage 3 ($N = 34,852$), and 6.6% as CKM stage 4 ($N = 12,387$). Comparisons of the baseline characteristics between participants included or excluded in the main analyses are displayed in Supplementary Table 3. As CKM stages progressed, participants were more likely to be older, male, less educated, of lower socioeconomic status (SES), engage in fewer moderate-to-high physical activities, have weaker social connections, and experience poorer sleep quality (all $p < 0.05$). Additionally, individuals at higher CKM stages exhibited increased incidence rates of mental disorders, dementia, and mortality during the follow-up period (all $p < 0.05$).

3.2. Multi-state analyses

The numbers and percentages of participants transitioning from a healthy status to mental disorders, dementia, or death are illustrated in Figure 2a. Results from the multi-state model (Figure 2b) present the HRs for the impact of CKM staging on the temporal progression from a healthy state to the onset of mental disorders or dementia via Path

A or Path B.

In Path A, each one-stage increment in CKM stage was associated with a higher risk of developing mental disorders (Healthy → Mental Disorders: HR=1.24, 95% CI: [1.22, 1.26]), followed by a higher incidence of dementia (Mental Disorders → Dementia: HR=1.38, 95% CI: [1.21, 1.58]). In Path B, each one-stage increment in CKM stage was associated with an increased risk of directly developing dementia (Healthy → Dementia: HR=1.27, 95% CI: [1.21, 1.33]) but not subsequent mental disorders.

When treating CKM stage as a categorical variable, each CKM stage, compared with CKM stage 0, was significantly associated with an increased risk of incident mental disorders in Path A (Healthy → Mental Disorders), with HRs ranging from 1.30 to 2.57 across CKM stages 1 to 4. In Path B (Healthy → Dementia), only advanced CKM stages (CKM stage 3/4) were significantly associated with an increased risk of incident dementia.

We repeated the multi-state analyses using the Multistate Markov model and found consistent results (Supplementary Table 4).

3.3. Competing risk analyses

Results from the multi-state models demonstrated that CKM was associated with the transition from a healthy status, but not from dementia, to mental disorders. Therefore, we employed Fine-Gray sub-distribution hazard models to assess the relationship between CKM stage and the incidence of pre-dementia mental disorders, including specific subtypes, whilst accounting for death as a competing risk (Table 2). Survival curves of different pre-dementia mental disorders by CKM stage are shown in Figure 3. Each one-stage increment in CKM stage was associated with an elevated risk of developing any pre-dementia mental disorders (Model 2, sHR = 1.32, 95% CI: [1.30, 1.35]). Using CKM stages 0 to 3 as reference groups, higher CKM stages were associated with increased hazards of incidence depressive and anxiety disorders.

Conversely, the multi-state model results showed that CKM was associated with transitions both from a healthy status to dementia and from mental disorders to subsequent dementia. Therefore, we conducted Fine-Gray sub-distribution hazard

models in participants irrespective of preexisted mental disorders, assessing risks for all-cause dementia, Alzheimer's disease, and vascular dementia (Table 3). Survival curves of different types of dementia by CKM stage are shown in Figure 3. Whilst each one-stage increment in CKM stage was associated with an elevated risk of developing all-cause and specific types of dementia, significant associations were primarily observed in the advanced CKM stages (CKM stage 3/4).

3.4. Metabolomics analyses

Out of 251 metabolites examined, only 11 showed no significant association with CKM stage (Figure 4a). Among the significant associations, 118 metabolites—including creatinine and total triglycerides—were inversely associated with a worsening CKM whereas 122 metabolites—such as total lipids in HDL and degree of unsaturation—were positively associated (Supplementary Data 5). In the training set, the elastic net model selected 155 of these 240 significant metabolites for inclusion in the CKM metabolomic signature (Figure 4b). The CKM metabolomic signature demonstrated a positive correlation with the increasing CKM stages in training set, testing set, and combined data (all $p < 0.001$; Figure 4c). Furthermore, higher CKM metabolomic signature score was significantly associated with an increased risk of both pre-dementia mental disorders (HR per SD = 1.35, 95% CI: 1.29–1.41) and dementia (HR per SD = 1.19, 95% CI: 1.18–1.21) (Figure 4d). Mediation analysis revealed that the CKM metabolomic signature accounted for 34.9 % ($p < 0.001$) of the association between CKM stages and incident pre-dementia mental disorders, and 8.1% ($p = 0.04$) of the association between CKM stages and incident dementia (Figure 4d).

3.5. Sensitivity analyses

We conducted sensitivity analyses to examine the associations between CKM stage and both any pre-dementia mental disorders and dementia, as well as specific subtypes. These analyses used multiple imputation with a Markov Chain Monte Carlo approach and complete-case analyses, and the results were consistent with the primary findings (Supplementary Data 6-7). Similar sHRs for CKM syndrome across multiple outcomes were observed after further adjustment for medication use (Supplementary Data 8). In addition, we repeated the main analyses after excluding participants with ≤ 5

years and ≤ 10 years of follow-up. Although confidence intervals were wider in the ≤ 10 -year exclusion analysis, vascular dementia remained the outcome most strongly associated with CKM stage (Supplementary Data 9-10).

4. Discussion

The present prospective study of 375,203 midlife and older adults found that CKM syndrome exhibited a stepwise association with the progression of mental disorders, especially in depressive and anxiety disorders, during the pre-dementia phase. Notably, only advanced CKM syndrome (stages 3 or 4) was associated with dementia incidence, irrespective of any pre-existing mental disorders. We developed a CKM-related metabolomic signature, and we found that it was associated with both pre-dementia mental disorders and dementia and that it partially mediated the associations between CKM syndrome and neuropsychiatric outcomes.

Using data from the UK Biobank, we found that 86.6% of participants met the criteria for CKM syndrome (stage 1 or higher), with the majority classified as CKM stage 2 (49.1%). The prevalence reported in our findings aligns with a recent study examining the prevalence of CKM stages across different age, sex, and race groups in U.S. adults older than 45 years of age, highlighting widespread poor CKM health²⁰. Notably, our findings are consistent with a recent study showing that females, compared with males, had a lower prevalence of CKM stages 3 and 4⁴⁸. This trend also aligns with previous studies demonstrating a higher risk of CKD and CVD among males^{49,50}. Given the high prevalence of CKM syndrome, healthcare strategies focused on improving CKM health are urgently needed¹⁹. Adopting the definition developed by the American Heart Association (AHA) may help clarify these adverse health interactions and provide a framework for staging, screening, risk stratification, prevention, and treatment^{22,51}.

We found that CKM syndrome is primarily associated with mental disorders during pre-dementia phase. Specifically, CKM stages exhibited a stepwise association with higher hazards of developing stress-related disorders, including depression and anxiety, compared to the previous stage. These findings are consistent with previous studies⁵²⁻⁵⁵, such as a recent study demonstrating that metabolic dysregulations were

linked to a long-term (21-year) risk of depression, anxiety, and stress-related disorders⁵². Additionally, our prior research found that metabolic multimorbidity was associated with the presence and incidence of affective subsyndromes (primarily comprising depression and anxiety symptoms) among pre-dementia older adults⁵⁵. Nevertheless, we cannot conclude that there is no association between CKM syndrome and other types of mental disorders, such as psychotic and bipolar disorders. We observed attenuated associations between CKM syndrome and these conditions. One possible explanation is their lower incidence (fewer than 700 cases during the follow-up period). Previous literature has found that severe mental illnesses, including psychotic and bipolar disorders, are associated with an increased risk of metabolic dysfunction, cardiovascular disease (CVD), and CVD-related mortality^{56,57}. However, there is limited evidence on how kidney, metabolic, and cardiovascular factors specifically impact psychotic and bipolar disorders.

On the other hand, both mood disturbances and mental disorders prior to dementia onset are significant neuropsychiatric manifestations of cognitive impairment and dementia^{1,58}. Several mental disorders, including psychotic, bipolar, depressive, and anxiety disorders, are considered prodromal to dementia and were included in this study^{9,59}. Building on previous research, although we found that CKM syndrome significantly increased the risk of developing AD or VaD only at subclinical or clinical CVD stages (advanced stages), the observed relationship between early CKM stages and pre-dementia mental disorders offers insights on the holistic disease transitioning flowchart from CKM syndrome to mental disorders, and eventually to dementia.

Our findings indicate that more advanced CKM stages were predominantly associated with vascular dementia, suggesting an underlying cerebrovascular pathology. Advanced CKM stages are characterized by subclinical/clinical CVD and severe kidney dysfunction, both of which have been widely reported to be associated with cerebral vascular damages, including, cerebral small vessel disease⁶⁰ (e.g., white matter hyperintensities), cerebral hypoperfusion, and microvascular injury⁶¹. However, because CKM syndrome comprises multiple heterogeneous components, future studies are warranted to examine how different CKM stages—or distinct phenotypic subtypes

within the same stage—are differentially associated with neurodegenerative processes and cerebrovascular damage.

The associations of metabolic syndrome, CKD, and CVD, as well as their comorbidity, with mental disorders and dementia have been reported in the geriatric population^{12–17}. Rather than merely considering additive combinations of different chronic diseases, the conceptual definition of CKM syndrome allows for a progressively layered investigation of its impact on neuropsychiatric outcomes. By including both midlife and older adults, we emphasize the critical need for regular monitoring and targeted interventions for pre-dementia mental disorders at each stage of CKM syndrome. Adopting this comprehensive CKM staging framework could enhance the effective management of CKM syndrome progression and potentially mitigate the onset of geriatric neuropsychiatric disturbances.

We constructed the CKM metabolomic signature using 155 metabolites to reflect CKM stage progression. Our analyses revealed that this metabolomic signature mediated a substantial proportion of the association between CKM stages and neuropsychiatric and neurocognitive outcomes: 34.9% for mental disorders versus 8.1% for dementia. These differential mediation effects suggest distinct pathological mechanisms—metabolic dysregulation appears to be a prominent, modifiable pathway contributing to mood impairments, whereas dementia risk likely involves additional factors (e.g., vascular injury, amyloid pathology) beyond metabolic disturbances. These findings highlight the potential for stage-specific interventions. Whilst early metabolic optimization may be more effective for psychiatric symptom prevention, comprehensive risk factor management may be necessary to mitigate dementia risk. Further studies are needed to clarify and understand the underlying mechanisms of the unexplained portion.

The strengths of this study include the use of a community-based cohort with a large sample size encompassing both midlife and older adults, as well as a long follow-up period. Another key strength is the application of 2 multi-state models (the Cox model and Multistate Markov model) which allowed for a more precise assessment of the impact of CKM stages on each transition, and to demonstrate the robustness of study

findings. However, several limitations should be considered. Given the observational nature of our study, the causality between CKM syndrome, mental disorders, and dementia warrant further investigation. Additionally, the UK Biobank cohort is more likely to be from less deprived areas and have fewer health conditions, which may introduce selection bias⁶². As a result, the effects of CKM risk could be more pronounced in more representative cohorts. Although we adjusted for a range of known potential confounders, residual confounding due to unmeasured factors, such as the use of statins or antidepressants, may still be present. Sensitivity analyses excluding participants with less than 5 or 10 years of follow-up yielded results consistent with the main findings; however, the possibility of reverse causation cannot be completely excluded. Medical condition data were obtained from self-report and medical records, which carry the risk of misreporting or missed diagnoses⁶³. The future availability of comprehensive primary care data in the UK Biobank may help validate medical condition information. Due to the absence of imaging data or cardiac biomarkers, we used the FRS as a proxy for subclinical cardiovascular disease. However, this surrogate measure may not fully capture true subclinical CVD, potentially resulting in misclassification of CKM stage 3. Although we separated the data into training and testing sets and developed the CKM metabolomic signature using 10-fold cross-validation to reduce the risk of overfitting and to ensure internal validity, the absence of external validation in independent cohorts may limit the generalizability and reproducibility of our findings in other populations. Lastly, our study was limited to participants of European ancestry, and further research is needed to determine whether these findings are generalizable to other populations.

5. Conclusions

Given the high prevalence of CKM syndrome, our findings underscore the need of managing CKM progression to reduce geriatric neuropsychiatric disturbances. The stage-specific associations between CKM syndrome, pre-dementia mental disorders, and dementia could help to tailor screening and management strategies across CKM stages. Furthermore, the mediating role of the CKM-related metabolomic signature suggests that metabolic modification may represent a potential avenue for the

prevention of psychiatric symptoms and cognitive decline.

6. Data availability

The corresponding UK Biobank field IDs for all variables are provided in the Supplementary Materials. Researchers registered with the UK Biobank may apply for access to the database by submitting an application that includes a summary of the research plan, the required data fields, details of any new data or variables to be generated, and payment to cover the incremental costs of servicing the application (<https://www.ukbiobank.ac.uk/enable-your-research/apply-for-access>). Source data underlying Figs.4 is provided in Supplementary Data 5.

7. Code availability

All analyses were performed using publicly available R software (R version 4.4.2 (<https://www.r-project.org/>)). The code used in this study can be made available from the corresponding author on reasonable request.

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Competing interests

The authors have no interests to disclose.

Author contribution

X X conceptualized the study. XH Z, ZY L, and WH Z performed the data analysis and wrote the first draft. C C, SH H, HW J, HR Z, and X X reviewed the biostatistical results and revised the manuscript, conceived the study and methodology, and interpreted the data. ZY L and X X were responsible for funding acquisition, data curation, and investigation. All authors revised the manuscript and provided critical revisions to the manuscript. All authors were responsible for reviewing and editing the manuscript.

Figure caption and legend:**Figure 1. Study flowchart.**

The flowchart illustrates the selection of participants from the initial study population to the final analytic cohorts. For the main analyses, participants with incomplete CKM data, dementia, major diseases, or mental disorders at baseline were excluded, resulting in a final sample of 375,203 participants. For the metabolomic analyses, additional exclusions were made for missing plasma metabolomic data, yielding a final analytic cohort of 188,018 participants. CKM denotes cardiovascular–kidney–metabolic syndrome.

Figure 2. Multi-state model in all transition patterns from baseline to mental disorders or dementia.

(a) Numbers (percentages) of participants in all transition patterns from baseline to psychiatric disorders or dementia and subsequently to death. The red lines represent Path A: from healthy status to psychiatric disorder to then to dementia while the blue lines represent Path B: from healthy status to dementia to then to psychiatric disorder. Participants who were diagnosed with psychiatric disorders and dementia at same time were excluded in multi-state model. (b) Hazard ratios (95% CIs) for Path A and Path B. Model was adjusted for age, sex, education, alcohol assumption, socioeconomic status, physical activity, social connection, and sleep duration. HR, hazard ratio; CI, confidence interval.

Figure 3. Survival curves for mental disorders and dementia across Cardiovascular–Kidney–Metabolic syndrome stages.

Solid lines represent estimated survival functions, and shaded areas indicate 95% confidence intervals. Among 375,203 participants, 20,861 developed any mental disorders, 685 developed psychotic disorders, 637 developed bipolar disorders, 17,150 developed depressive disorders, 13,898 developed anxiety disorders, 4,421 developed all-caused dementia, 1,895 developed Alzheimer's disease and 994 developed Vascular dementia during follow-up.

Figure 4. Construction of the CKM metabolomic signature and mediation analyses.

(a) Volcano plot showing associations between 251 NMR-based plasma metabolites and CKM stages. Associations were assessed using multivariable linear regression with two-sided tests, adjusting for age, sex, race, education, alcohol consumption, socioeconomic status, physical activity, social connection, and sleep quality ($N = 188,018$). Orange (positive) and blue (negative) dots indicate statistically significant associations, while grey dots denote non-significant associations. The horizontal dashed line represents the Bonferroni-corrected significance threshold for multiple comparisons ($p < 3.31 \times 10^{-4}$). (b) Feature selection using elastic net regression within a tenfold cross-validation framework in the training set ($N = 150,414$; 80% of the metabolomic cohort). The model was trained to predict CKM stages using the 240 metabolites significantly associated with CKM stages. (c) Correlations between the CKM metabolomic signature (comprising 155 selected metabolites) and CKM stages

in the training set ($N = 150,414$), testing set ($N = 37,604$), and the combined dataset ($N = 188,018$). Pearson correlation coefficients were calculated using two-sided tests. (d) Mediation analyses examining the role of the CKM metabolomic signature in the associations between CKM stage, pre-dementia mental disorders, and dementia ($N = 188,018$). The proportion mediated was calculated as the proportion of the CKM-stage coefficient attenuated after inclusion of the CKM metabolomic signature, with 95% confidence intervals estimated using non-parametric bootstrapping. Abbreviations. CI confidence interval, HR hazard ratio, CKM cardiovascular–kidney–metabolic syndrome, NMR nuclear magnetic resonance.

Table 1 Sample Characteristics by Cardiovascular-Kidney-Metabolic syndrome stages

Characteristic	Overall(n=375203)	Cardiovascular-Kidney-Metabolic syndrome stage					p value
		stage0(n=48955)	stage1(n=41750)	stage2(n=184331)	stage3(n=79775)	stage4(n=20392)	
Demographic/Covariates							
Age, years (mean±SD)	56.53±8.09	52.05±7.70	52.76±7.79	55.36±7.74	62.59±5.03	61.87±6.05	<0.001
Sex, female (n, %)	201994 (53.8)	36863 (75.3)	29446 (70.5)	114856 (62.3)	14624 (18.3)	6205 (30.4)	<0.001
Race (n, %)							<0.001
White	358342 (95.8)	47206 (96.7)	39569 (95.1)	175223 (95.4)	76919 (96.8)	19425 (95.6)	
Black	7193 (1.9)	735 (1.5)	1313 (3.2)	3879 (2.1)	958 (1.2)	308 (1.5)	
Asian	8364 (2.2)	894 (1.8)	732 (1.8)	4583 (2.5)	1572 (2.0)	583 (2.9)	
Education, high (n, %)	176976 (47.7)	29164 (59.9)	21361 (51.6)	86695 (47.5)	32847 (41.7)	6909 (34.5)	<0.001
Alcohol assumption (n, %)							<0.001
Current	346271 (92.4)	45912 (93.9)	38928 (93.4)	169494 (92.1)	74031 (92.9)	17906 (88.0)	
Former drinker	12479 (3.3)	1335 (2.7)	1169 (2.8)	5882 (3.2)	2846 (3.6)	1247 (6.1)	
Never	16005 (4.3)	1669 (3.4)	1603 (3.8)	8710 (4.7)	2799 (3.5)	1194 (5.9)	
SES ¹ (median, IQR)	12.52 [7.02, 22.85]	10.93 [6.28, 19.88]	12.28 [6.9, 22.36]	12.66 [7.08, 22.97]	12.66 [7.12, 23.23]	15.68 [8.46, 29.34]	<0.001
Physical activity, high (n, %)	294225 (80.1)	40744 (84.0)	33333 (81.0)	143375 (79.5)	61893 (79.7)	14880 (76.0)	<0.001
Social connection, high (n, %)	350461 (94.0)	46246 (95.0)	39380 (94.8)	171902 (93.9)	74586 (94.2)	18347 (91.0)	<0.001
Sleep quality, good (n, %)	275499 (73.9)	38172 (78.2)	30909 (74.4)	133476 (72.9)	59264 (74.7)	13678 (67.8)	<0.001
Incident mental disorders							
Any (n, %)	20861 (5.6)	1913 (3.9)	2123 (5.1)	10895 (5.9)	4124 (5.2)	1806 (8.9)	<0.001
Bipolar disorders (n, %)	637 (0.2)	65 (0.1)	56 (0.1)	341 (0.2)	137 (0.2)	38 (0.2)	0.041
Psychotic disorders (n, %)	685 (0.2)	56 (0.1)	65 (0.2)	333 (0.2)	191 (0.2)	40 (0.2)	<0.001
Depressive disorders (n, %)	17150 (4.6)	1566 (3.2)	1762 (4.2)	9018 (4.9)	3283 (4.1)	1521 (7.5)	<0.001
Anxiety disorders (n, %)	13898 (3.7)	1372 (2.8)	1381 (3.3)	7330 (4.0)	2717 (3.4)	1098 (5.4)	<0.001
Incident dementia							
All-cause dementia (n, %)	4421 (1.2)	220 (0.4)	191 (0.5)	1474 (0.8)	1766 (2.2)	770 (3.8)	<0.001
Alzheimer’s disease (n, %)	1895 (0.5)	115 (0.2)	79 (0.2)	669 (0.4)	775 (1.0)	257 (1.3)	
Vascular dementia (n, %)	994 (0.3)	24 (0.0)	24 (0.1)	232 (0.1)	438 (0.5)	276 (1.4)	
Incident death (n, %)	31220 (8.3)	1677 (3.4)	1618 (3.9)	11035 (6.0)	12127 (15.2)	4763 (23.4)	<0.001

Note: χ^2 tests or analysis of variance (ANOVA) were used, as appropriate, to compare participant characteristics across CKM stages. SES denotes socioeconomic status, assessed using the Townsend Deprivation Index. High education refers to attainment of higher secondary education or above. High physical activity indicates engagement in moderate or vigorous physical activity for at least 10 minutes on two or more days per week. High social connection is defined as not often feeling lonely or living with others. Good sleep quality is defined as sleep duration between 7 and 9 hours. Abbreviations. IQR, inter-quartile range; SD, standard deviation.

Table 2 Associations of Cardiovascular-Kidney-Metabolic syndrome stages with any and different types of pre-dementia mental disorders

	Pre-dementia mental disorders No. Cases=20,861		Psychotic disorders No. Cases=685		Bipolar disorders No. Cases=637		Depressive disorders No. Cases=17,150		Anxiety disorders No. Cases=13,898	
	Subdistribution hazard ratio (95% CI)									
	model1	model2	model1	model2	model1	model2	model1	model2	model1	model2
Reference = CKM stage0										
CKM stage1	1.35 (1.27, 1.44)***	1.28 (1.20, 1.36)***	1.30 (0.91, 1.86)	1.21 (0.81, 1.78)	1.04 (0.73, 1.50)	1.04 (0.72, 1.52)	1.38 (1.29, 1.47)***	1.30 (1.21, 1.39)***	1.21 (1.13, 1.31)***	1.14 (1.05, 1.23)**
CKM stage2	1.69 (1.61, 1.78)***	1.50 (1.43, 1.58)***	1.51 (1.13, 2.01)*	1.37 (1.00, 1.87)*	1.54 (1.17, 2.03)**	1.39 (1.04, 1.85)*	1.74 (1.65, 1.84)***	1.53 (1.45, 1.62)***	1.51 (1.42, 1.60)***	1.35 (1.27, 1.43)***
CKM stage3	2.04 (1.92, 2.18)***	1.71 (1.60, 1.83)***	1.91 (1.36, 2.68)**	1.55 (1.07, 2.25)*	1.75 (1.24, 2.48)***	1.54 (1.07, 2.21)*	2.08 (1.94, 2.23)***	1.73 (1.60, 1.86)***	1.71 (1.59, 1.85)***	1.46 (1.35, 1.58)***
CKM stage4	3.41 (3.18, 3.65)***	2.46 (2.28, 2.65)***	1.66 (1.08, 2.56)*	1.15 (0.72, 1.83)	1.90 (1.23, 2.91)**	1.24 (0.77, 1.99)	3.64 (3.37, 3.92)***	2.57 (2.37, 2.79)***	2.61 (2.40, 2.84)***	1.96 (1.79, 2.15)***
Reference = CKM stage1										
CKM stage2	1.25 (1.20, 1.31)***	1.18 (1.12, 1.24)***	1.16 (0.89, 1.52)	1.13 (0.85, 1.52)	1.48 (1.11, 1.97)***	1.33 (0.99, 1.79)	1.26 (1.20, 1.33)***	1.18 (1.12, 1.25)***	1.25 (1.18, 1.32)***	1.18 (1.11, 1.26)***
CKM stage3	1.51 (1.42, 1.61)***	1.34 (1.26, 1.43)***	1.47 (1.06, 2.03)**	1.29 (0.91, 1.83)	1.68 (1.18, 2.39)***	1.47 (1.02, 2.13)*	1.51 (1.41, 1.61)***	1.33 (1.24, 1.43)***	1.41 (1.31, 1.52)***	1.28 (1.19, 1.39)***
CKM stage4	2.52 (2.36, 2.70)***	1.93 (1.79, 2.07)***	1.28 (0.84, 1.94)	0.95 (0.61, 1.49)	1.82 (1.17, 2.80)**	1.19 (0.73, 1.91)	2.64 (2.45, 2.84)***	1.98 (1.83, 2.14)***	2.15 (1.98, 2.34)***	1.73 (1.58, 1.89)***
Reference = CKM stage2										
CKM stage3	1.21 (1.16, 1.26)***	1.14 (1.09, 1.19)***	1.27 (1.02, 1.57)*	1.14 (0.90, 1.43)	1.14 (0.90, 1.44)	1.11 (0.86, 1.42)	1.19 (1.14, 1.25)***	1.13 (1.07, 1.18)***	1.13 (1.08, 1.19)***	1.09 (1.03, 1.15)***
CKM stage4	2.01 (1.91, 2.12)***	1.64 (1.55, 1.73)***	1.10 (0.78, 1.55)	0.84 (0.58, 1.21)	1.23 (0.87, 1.74)	0.89 (0.60, 1.32)	2.09 (1.98, 2.22)***	1.67 (1.57, 1.78)***	1.73 (1.62, 1.85)***	1.46 (1.36, 1.57)***
Reference = CKM stage3										
CKM stage4	1.67 (1.58, 1.76)***	1.44 (1.35, 1.53)***	0.87 (0.62, 1.23)	0.74 (0.51, 1.07)	1.08 (0.75, 1.55)	0.80 (0.54, 1.21)	1.75 (1.65, 1.86)***	1.49 (1.39, 1.59)***	1.52 (1.42, 1.64)***	1.34 (1.25, 1.45)***
Continuous variable, per 1 increase in CKM stage	1.32 (1.30, 1.35)***	1.23 (1.21, 1.25)***	1.17 (1.07, 1.28)*	1.08 (0.98, 1.19)	1.21 (1.10, 1.33)***	1.11 (1.01, 1.23)*	1.34 (1.32, 1.37)***	1.24 (1.21, 1.26)***	1.25 (1.22, 1.27)***	1.17 (1.14, 1.19)***

Note: Model 1 was adjusted for age, sex, race. Model 2 was adjusted for age, sex, race, education, alcohol assumption, socioeconomic status, physical activity, social connection, and sleep quality. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Abbreviations. CKM, cardiovascular-kidney-metabolic syndrome; CI, confidence interval.

Table 3 Associations of Cardiovascular-Kidney-Metabolic syndrome stages with all-case and different types of dementia

	All-cause dementia No. Cases=4,421		Alzheimer's disease No. Cases=1,895		Vascular dementia No. Cases=994	
	Subdistribution hazard ratio (95% CI)					
	model1	model2	model1	model2	model1	model2
Reference = CKM stage0						
CKM stage1	0.91 (0.75, 1.11)	0.91 (0.74, 1.11)	0.71 (0.54, 0.95)*	0.69 (0.51, 0.93)*	1.04 (0.59, 1.83)	1.05 (0.58, 1.89)
CKM stage2	1.13 (0.98, 1.31)	1.07 (0.92, 1.24)	0.93 (0.76, 1.13)	0.86 (0.70, 1.06)	1.59 (1.04, 2.42)*	1.47 (0.95, 2.29)
CKM stage3	1.44 (1.24, 1.67)***	1.29 (1.10, 1.51)**	1.09 (0.89, 1.35)	0.98 (0.79, 1.22)	2.93 (1.92, 4.47)***	2.46 (1.58, 3.84)***
CKM stage4	2.72 (2.32, 3.18)***	2.29 (1.94, 2.71)***	1.45 (1.16, 1.83)**	1.25 (0.98, 1.59)	7.47 (4.87, 11.47)***	5.84 (3.72, 9.16)***
Reference = CKM stage1						
CKM stage2	1.24 (1.07, 1.44)**	1.18 (1.01, 1.38)***	1.31 (1.03, 1.65)*	1.24 (0.97, 1.58)	1.53 (1.00, 2.33)*	1.40 (0.91, 2.15)
CKM stage3	1.57 (1.35, 1.84)***	1.42 (1.21, 1.67)***	1.53 (1.20, 1.95)*	1.41 (1.10, 1.82)***	2.81 (1.84, 4.29)***	2.33 (1.51, 3.61)***
CKM stage4	2.98 (2.53, 3.51)***	2.53 (2.13, 3.00)***	2.04 (1.57, 2.65)***	1.80 (1.37, 2.36)***	7.17 (4.68, 10.99)***	5.54 (3.57, 8.61)***
Reference = CKM stage2						
CKM stage3	1.27 (1.17, 1.38)***	1.20 (1.10, 1.31)***	1.17 (1.03, 1.33)**	1.14 (0.99, 1.30)	1.83 (1.53, 2.22)***	1.67 (1.37, 2.04)***
CKM stage4	2.40 (2.18, 2.65)***	2.14 (1.93, 2.37)***	1.56 (1.34, 1.83)***	1.45 (1.23, 1.72)***	4.70 (3.86, 5.71)***	3.96 (3.21, 4.89)***
Reference = CKM stage3						
CKM stage4	1.89 (1.74, 2.06)***	1.78 (1.62, 1.95)*	1.33 (1.16, 1.53)***	1.27 (1.09, 1.50)***	2.55 (2.20, 2.97)***	2.37 (2.02, 2.79)***
Continuous variable, per 1 increase in CKM stage	1.35(1.30,1.40)***	1.27(1.22,1.32)***	1.14(1.08,1.21)***	1.10(1.03,1.16)***	1.99 (1.82, 2.16)***	1.82(1.66,2.00)***

Note: Model 1 was adjusted for age, sex, race. Model 2 was adjusted for age, sex, race, education, alcohol assumption, socioeconomic status, physical activity, social connection, and sleep quality. * p < 0.05, ** p<0.01, *** p<0.001. Abbreviations. CKM, cardiovascular-kidney-metabolic syndrome; CI, confidence interval.

ED SUM:

Zhao et al. assess links between cardiovascular–kidney–metabolic (CKM) syndrome, psychiatric disorders, and dementia, and the mediating role of a CKM-related metabolomic signature. Findings show that higher CKM stages are linked to psychiatric disorders, while only advanced stages are linked to dementia.

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502,462 participants from UK BioBank at baseline

Excluded 113,148 participants:
- Incomplete CKM data at baseline(n=111,297)
- Dementia, major disease and mental disorders at baseline(n=15,962)

Main Analyses

375,203 participants at baseline

CKM stage0
n=48,955

CKM stage1
n=41,750

CKM stage2
n=184,331

CKM stage3
n=79,775

CKM stage4
n=20,392

Excluded 187,185 participants:
- Without available plasma metabolomics data

Metabolomics Analyses

188,018 participants

CKM stage0
n=18,187

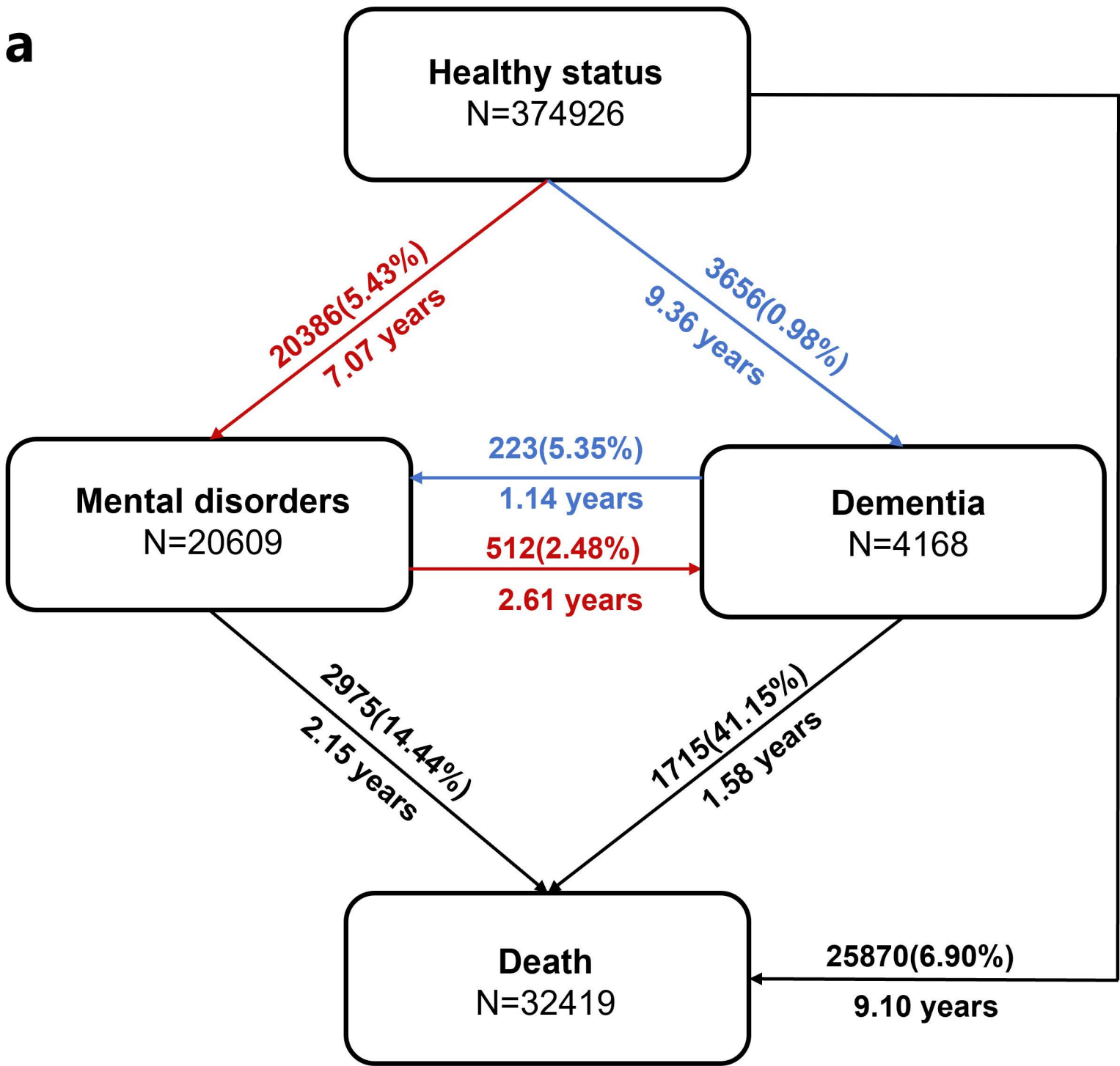
CKM stage1
n=14,332

CKM stage2
n=108,260

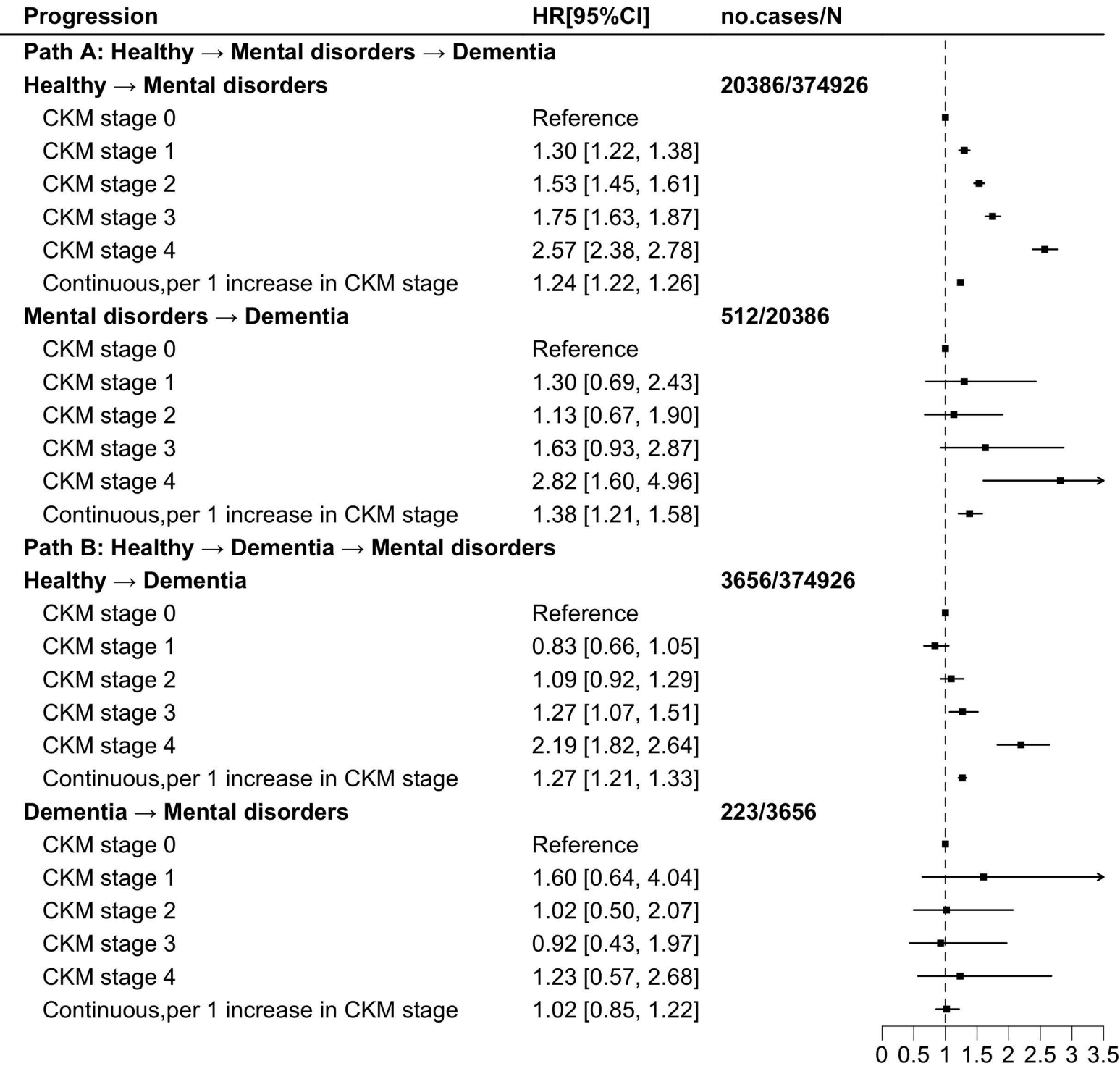
CKM stage3
n=34,852

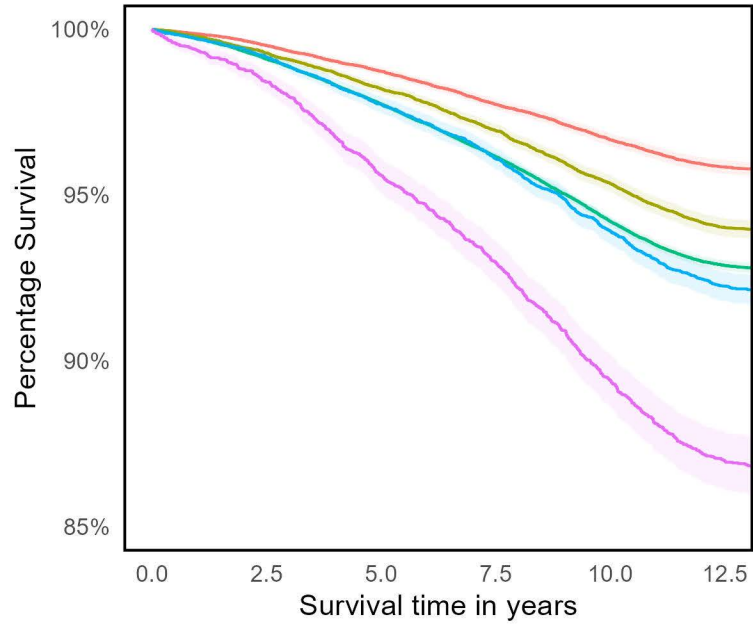
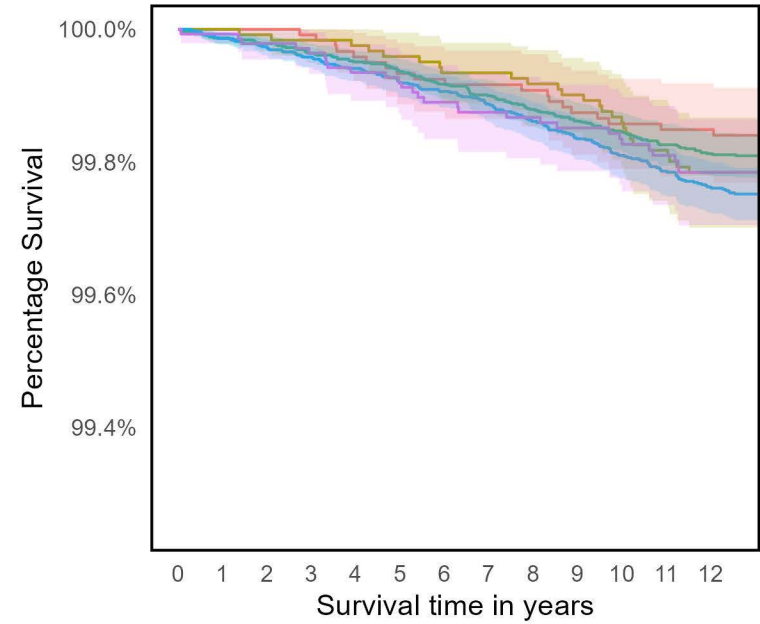
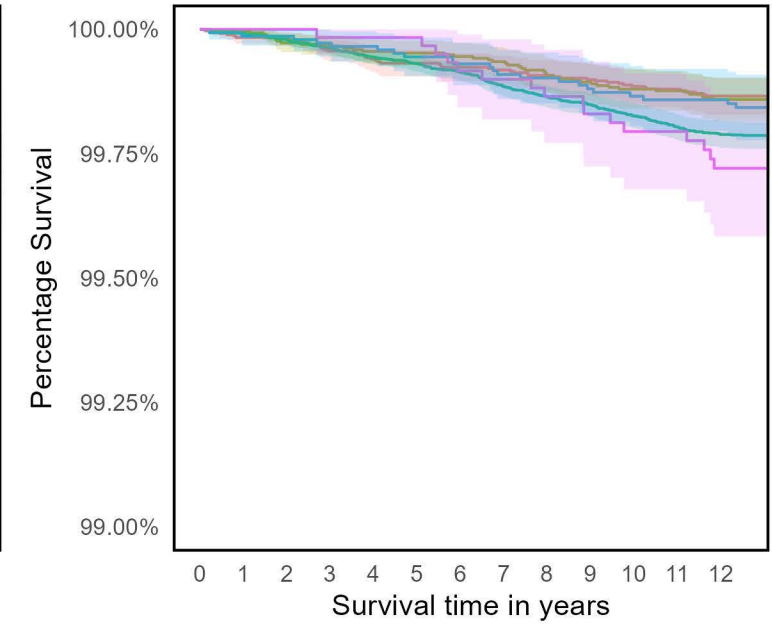
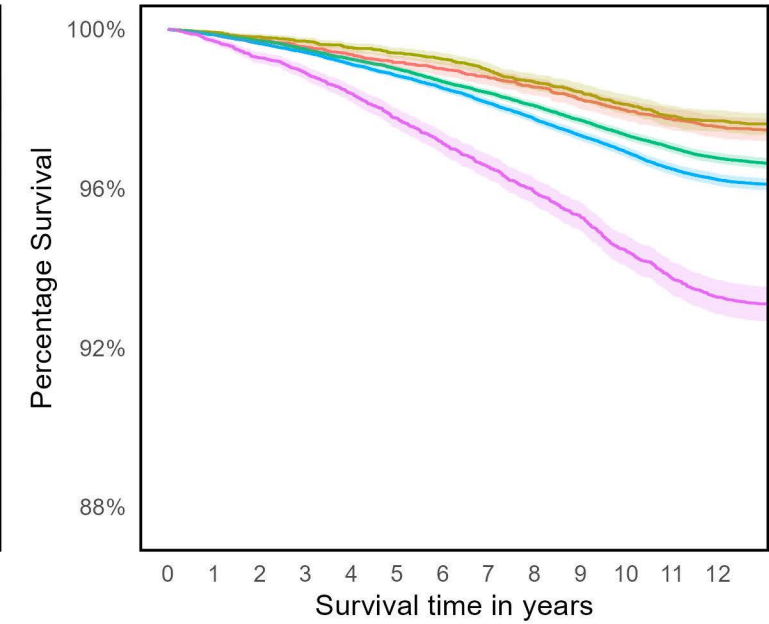
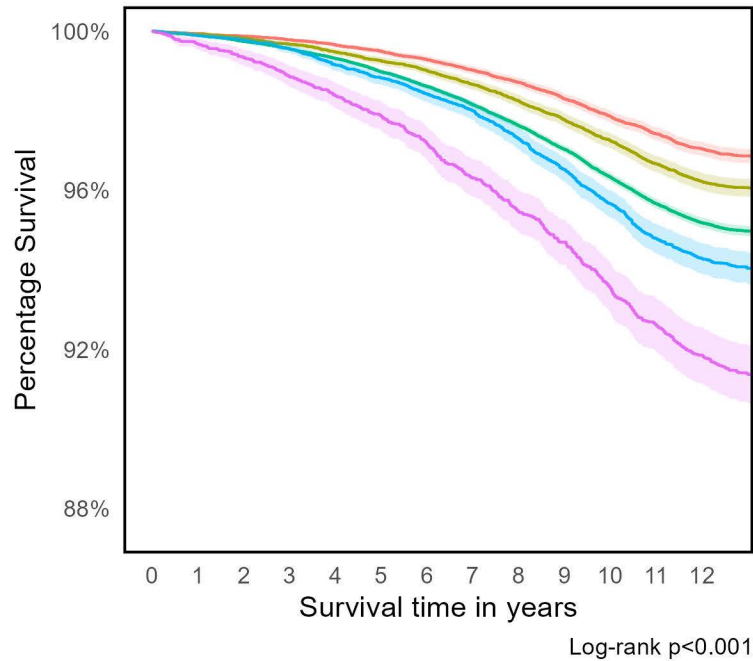
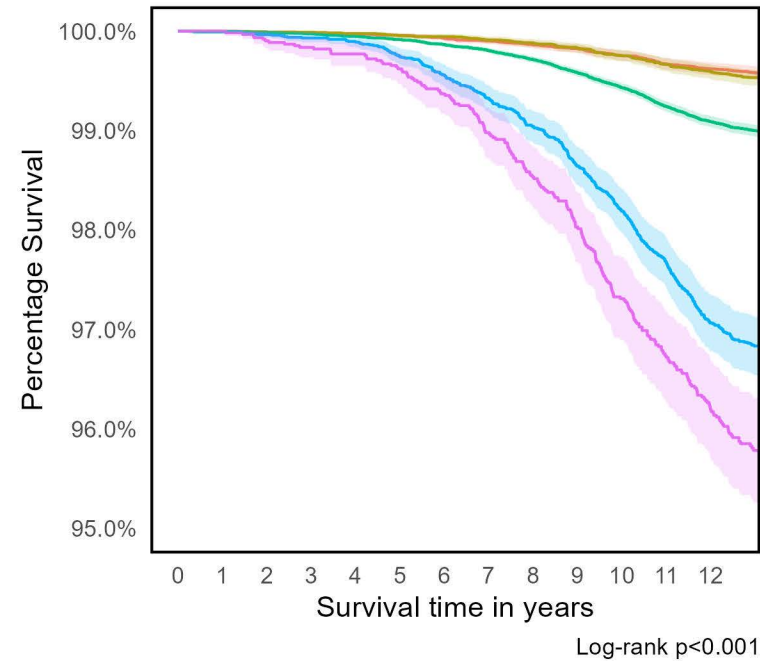
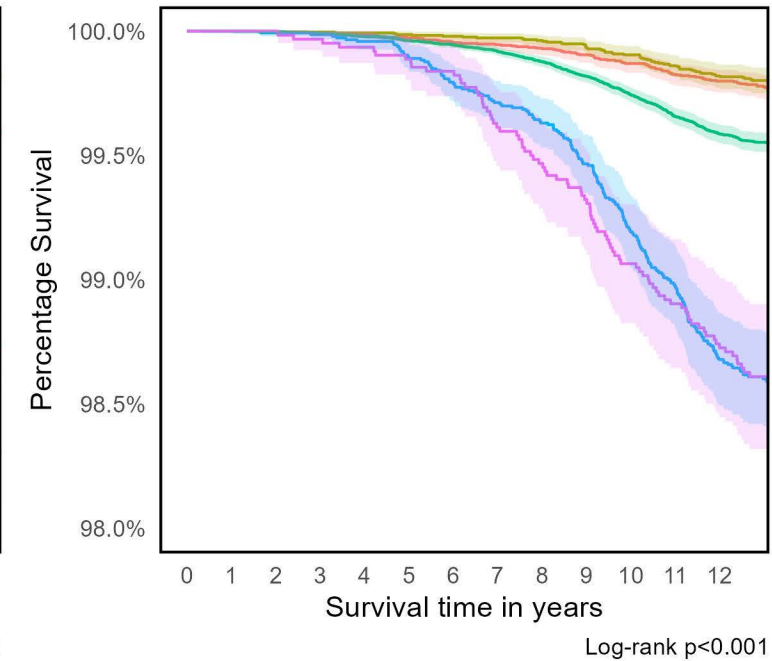
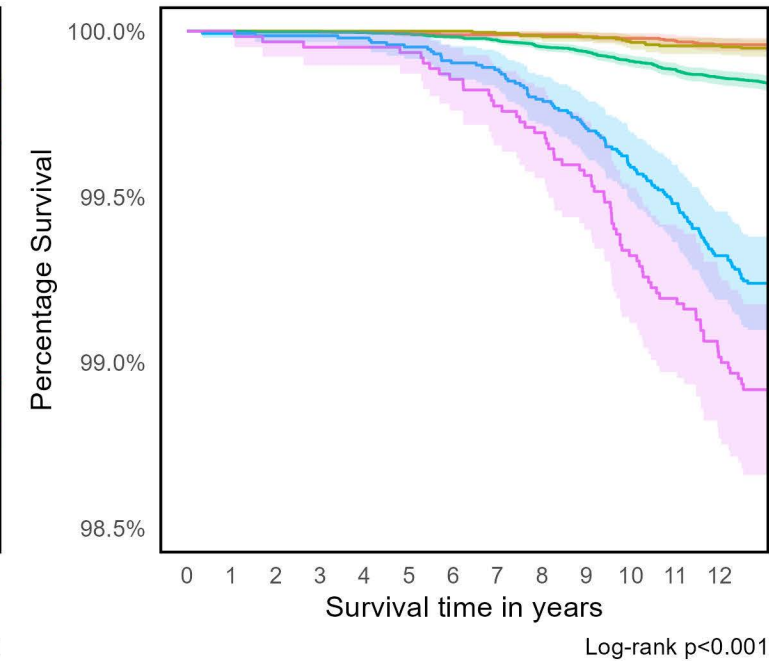
CKM stage4
n=12,387

a



b



Mental disorders**Psychotic disorders****Bipolar disorders****Depressive disorders****Anxiety disorders****All-cause dementia****Alzheimer's disease****Vascular dementia**

— Stage 0 — Stage 1 — Stage 2 — Stage 3 — Stage 4

